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Subject Code: DMITBA-01
Paper ID: 13441

GNIOT Institute of Management Studies – PGDM (Batch 2021-23)

(TRIMESTER – III) END TERM EXAMINATION, APRIL - 2022

STATISTICAL METHODS FOR DECISION MAKING USING R/PYTHON

[Time: 2 Hours 30 Minutes] ***[Maximum Marks: 100]***

INSTRUCTIONS:

(i) ***There are THREE sections in this question paper.***
(ii) ***Attempt questions from each section as per the directions.***
(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A

(Short Answer Type Questions, do not exceed 100 words)

Q.1. Attempt all parts of the following: ***(10×3=30)***

a) Difference between Correlation and Regression.
b) Explain Exploratory data analysis and types of exploratory data analysis.
c) Important differences between DDL and DML with example.
d) Import numpy or matplotlib module in two different ways.
e) Describe the terms with Python code.

I. Percentiles
II. Standard Derivations

f) List some of the most important SQL Commands
g) Inner join Vs Outer join exact differentiate with example
h) Explain the job of a Business Analyst with relevant example
i) List the Application of Clustering

SECTION-B

(Long Answer Type Questions)

Attempt all questions of the following: ***(4×10=40)***

Q.2.**(a)** Define Matplotlib histogram**.** Plot a histogram of student's age where data is given. With suitable title, and x and y label.

OR

(b) Define Matplotlib histogram.Print the generated data by importing packages and generating data by random radiant.

Q.3. **(a**) Give syntax of Nested if-else condition. Explain if X=10 and Y=20 print the greater, less or equal condition of variable while using nested if-else.

OR

(b) What is are decision tree? How does the Decision Tree algorithm work?

***Q.4.* (a)** State the understanding of four types of analytics of your data drive analytics maturity.

OR

(b) Describe logical operators' syntax, description with example.

Q.5. (a) List different built-in modules in Python. Explain any two.

OR

(b) We have registered the speed of 13 cars. What is the average, the middle, or the most common speed value? Speed = [99,86,87,88,111,86,103,87,94,78,77,85,86] Calculate the mean in Python.

SECTION-C ***(2×15=30)***

Sports Analytics is a game changer when it comes to how professional games are played, especially how strategic decision making happens, which until recently was primarily done based on "gut feeling" or adherence to past traditions. NumPy forms a solid foundation for a large set of Python packages which provide higher level functions related to data analytics, machine learning, and AI algorithms. These packages are widely deployed to gain real-time insights that help in decision making for game-changing outcomes, both on field as well as to draw inferences and drive business around the game of cricket. Finding out the hidden parameters, patterns, and attributes that lead to the outcome of a cricket match helps the stakeholders to take notice of game insights that are otherwise hidden in numbers and statistics.

Q.6. Attempt all parts of the following:

6(a) Numpy has been used for various kinds of cricket related sporting analytics describe some.

6(b) What are the key numpy capabilities utilized in game that hidden in number and statistics.